

FPGA-Based Digital Password Authentication System with Security Lockout Mechanism

Internship Project Documentation

Submitted By

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1. Project Overview

This project presents an FPGA-Based Digital Password Authentication System with a Security Lockout Mechanism. The system is designed to provide secure access control using a password-based authentication mechanism implemented on an Artix-7 FPGA. Users authenticate themselves by entering a 4-digit password through a 4x4 hexadecimal keypad.

The entered password is captured digit-by-digit by the password entry module and compared against the stored password using an XOR-based encryption scheme. A Finite State Machine (FSM) governs the authentication process by managing the PASS, FAIL and LOCK states. If the entered password is correct, the FPGA grants access, turns ON the green LED, and displays "PASS" on the MAX7219-based seven-segment display.

If an incorrect password is entered, the system denies access, turns ON the red LED, and displays "FAIL" on the seven-segment display for a short timeout period before allowing the next attempt. After three consecutive failed attempts, the FSM transitions into a LOCK state, displays "LOCK", and blocks further password entries until an authorized unlock key is pressed.

The project demonstrates the design and implementation of a secure digital authentication system using Verilog HDL, combining keypad scanning, finite state machine design, XOR-based password encryption, and seven-segment display driving on a single FPGA platform, making it suitable as a building block for locks, access panels and security-sensitive embedded applications.

2. Project Objectives

- Design a secure FPGA-based password authentication system.
- Authenticate users using a 4x4 hexadecimal keypad.
- Encrypt and verify the stored password using XOR-based encryption.
- Implement a Finite State Machine to manage PASS, FAIL and LOCK states.
- Provide visual feedback through LED indicators for grant, deny and lockout conditions.
- Display real-time authentication status on a MAX7219 seven-segment display.
- Implement a security lockout mechanism after repeated failed attempts.
- Demonstrate complete FPGA-based digital design using Verilog HDL.

3. Problem Statement

Conventional access control systems based on mechanical keys or simple toggle switches offer minimal security and provide no resistance against repeated unauthorized access attempts. Such systems typically lack any mechanism to detect or respond to brute-force password guessing. This project aims to address this gap by implementing an FPGA-based password authentication system that combines encrypted password verification with an automatic lockout mechanism, thereby improving resistance to unauthorized access attempts.

4. Existing System

Traditional electronic locks and access panels typically rely on a fixed password compared directly in plain form, with no encryption of the stored credential and no mechanism to limit repeated incorrect attempts. As a result, such systems remain vulnerable to brute-force attacks, and an intruder can make unlimited password attempts without triggering any protective response from the system.

5. Proposed System

The proposed system uses an Artix-7 FPGA as the central controller for password authentication. A 4x4 hexadecimal keypad is used to enter a 4-digit password, which is captured serially by the password entry module. The captured password is verified against a securely stored reference value using XOR-based encryption logic. A dedicated security finite state machine evaluates the comparison result: a correct password moves the system to the PASS state, while an incorrect password moves it to the FAIL state. The FSM also tracks the number of consecutive failed attempts, and after three consecutive failures, transitions to a LOCK state in which further password entries are ignored until an authorized unlock key is pressed.

6. Design and Architecture

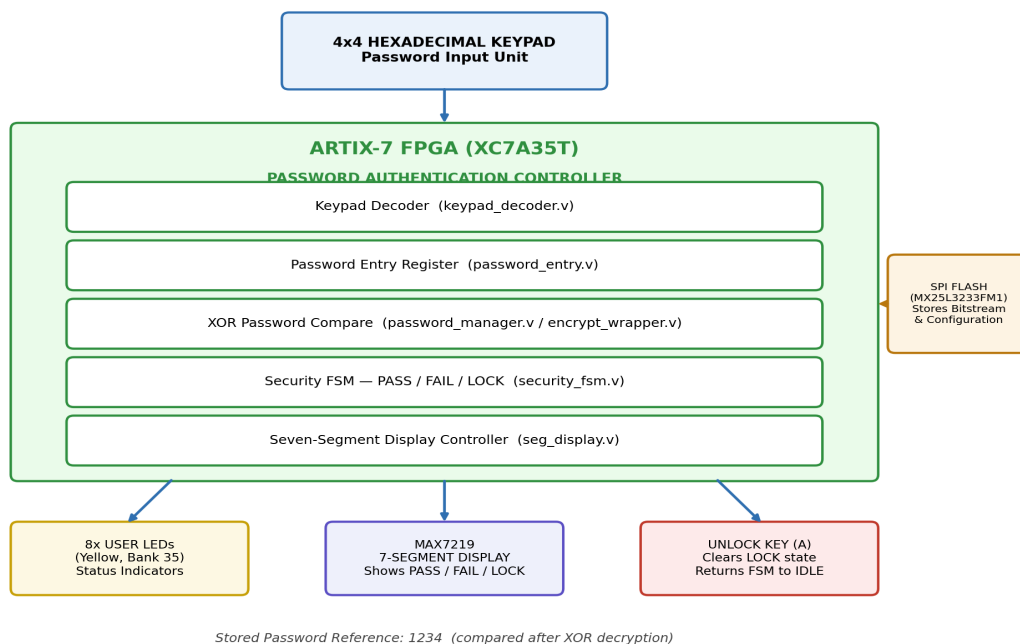


Figure 6.1: Block diagram of the FPGA-based digital password authentication system, showing the keypad input stage, the FPGA-internal authentication modules, and the LED/display output stage.

7. Hardware Components

- Artix-7 FPGA Development Board (Xilinx XC7A35T-FTG256-1)
- 4x4 Hexadecimal Keypad (keys 0-9, A-F)
- MAX7219-driven 4-digit Seven Segment Display Module
- 8x On-board User LEDs (Yellow, FPGA Bank 35) used as status indicators
- SPI Configuration Flash (MX25L3233FM1, 32Mb) for bitstream and program storage
- Onboard 24 MHz Clock Oscillator
- Push-button Reset Switch

Note: the development board provides 8 on-board user LEDs, all of the same yellow colour (FPGA Bank 35), as documented in the AT-STLN-ARTIX 7-001 Hardware Reference Manual. The authentication system therefore uses these yellow LEDs purely as status/activity indicators (reset, grant, deny, match, locked), while the actual PASS / FAIL / LOCK result is communicated to the user primarily through the MAX7219 seven-segment display.

8. Software Requirements

- Vivado Design Suite
- Verilog HDL
- Git (Version Control)

9. Working Principle

The user enters a 4-digit password through the hexadecimal keypad. The FPGA captures each digit and verifies the complete password, 1234, once all four digits have been entered. The reference password is stored in encrypted (XOR) form, and the program/bitstream itself is held in the onboard SPI configuration flash (MX25L3233FM1) so that the system boots automatically each time the board is powered on.

Correct password (1234):

- Access is granted by the security FSM.
- Seven-segment display shows "PASS".
- Status LEDs (yellow) indicate the grant/match condition.
- Failed-attempt counter is reset to zero.

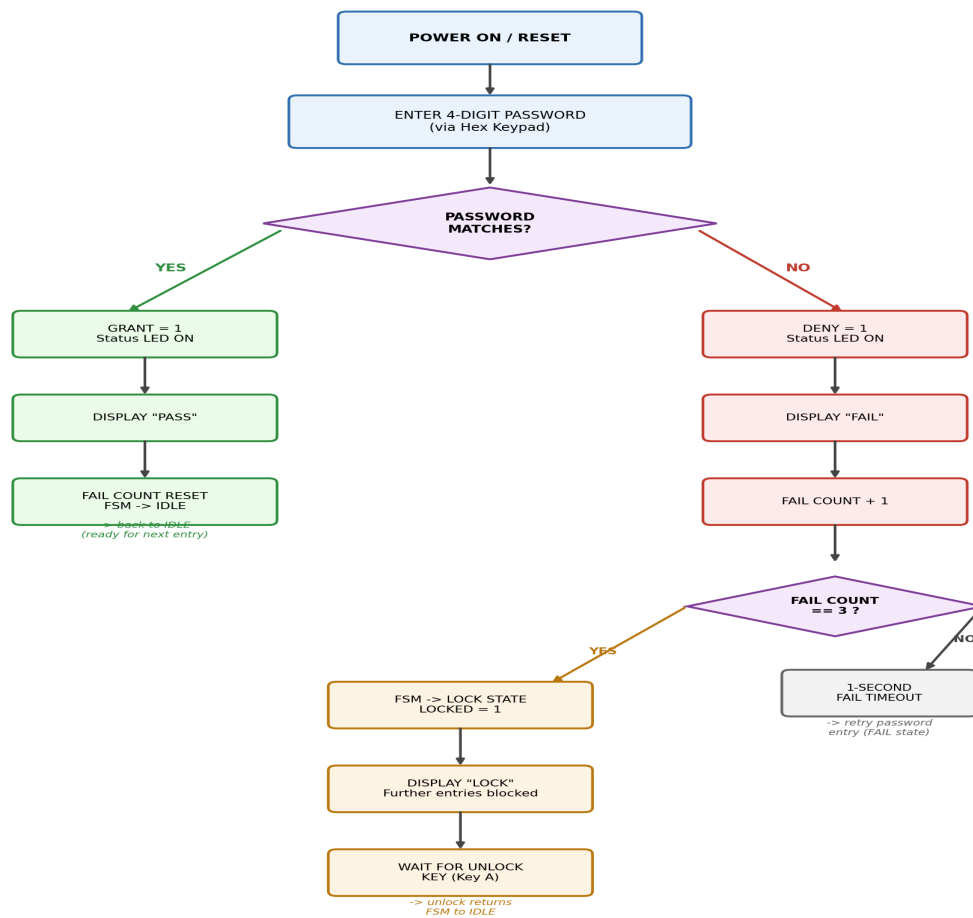
Incorrect password (any value other than 1234):

- Access is denied by the security FSM.
- Seven-segment display shows "FAIL".
- Status LEDs (yellow) indicate the deny condition.
- Failed-attempt counter increments by one.
- After a short timeout, the system returns to the idle state for a new attempt.

Lockout condition (three consecutive wrong attempts):

- After 3 consecutive incorrect password entries, the security FSM enters the LOCK state.
- Seven-segment display shows "LOCK".
- Further password entries are ignored and the system remains locked.
- Pressing key "A" on the hexadecimal keypad acts as the unlock key, clearing the LOCK state and returning the FSM to IDLE for a fresh password entry.

10. System Flow



Notes: Correct password (1234) -> PASS, status LED, FSM returns to IDLE.
Wrong password -> FAIL, status LED, 1s timeout, retry. Three consecutive
wrong attempts -> LOCK state until key "A" (unlock key) is pressed.

Figure 10.1: System flow of the password authentication process, including the PASS, FAIL and LOCK paths of the security finite state machine.

11. Implementation Approach

The system was implemented in Verilog HDL on Vivado Design Suite, targeting the Xilinx XC7A35T-FTG256-1 Artix-7 FPGA on the AT-STLN-ARTIX 7-001 board. The design follows a modular approach in which keypad scanning, password capture, encrypted comparison, finite state machine control and display driving are each implemented as independent, reusable Verilog modules and integrated under a single top-level module. The compiled bitstream is programmed into the onboard SPI configuration flash so that the authentication system loads automatically on every power-up.

11.1 Module Descriptions

1. keypad_decoder.v

Scans the 4x4 hexadecimal keypad inputs and generates a 4-bit key code along with a valid signal whenever a key press is detected.

2. password_entry.v

Captures each key press in sequence, builds the 16-bit (4-digit) entered password register, and generates a verify pulse once all four digits have been entered.

3. password_manager.v

Compares the entered password against the stored reference password and generates a match signal indicating whether the credentials are correct.

4. encrypt_wrapper.v

Implements XOR-based encryption of the password using a fixed key, providing a lightweight encryption layer for the stored credential.

5. security_fsm.v

Core finite state machine that manages the IDLE, SUCCESS (PASS), FAIL and LOCK states, tracks the consecutive failed-attempt counter, drives the grant/deny/locked outputs, and generates the display-state code used by the seven-segment display driver.

6. seg_display.v

Drives the MAX7219 seven-segment display module over a serial SPI-style interface to show "PASS", "FAIL" or "LOCK" depending on the current display state.

7. hsm_top.v

Top-level module that integrates the keypad decoder, password entry register, password comparator, security FSM and display driver into the complete authentication system, and maps internal status signals to the board LEDs.

11.2 Build Instructions

1. Open Vivado Design Suite.
2. Create a new RTL project.
3. Add all Verilog source files (hsm_top, keypad_decoder, password_entry, password_manager, encrypt_wrapper, security_fsm, seg_display).
4. Add the XDC constraints file for pin and clock assignments.

5. Run Synthesis.
6. Run Implementation.
7. Generate Bitstream.
8. Program the FPGA, then write the bitstream to the onboard SPI configuration flash (MX25L3233FM1) using Vivado's Add Configuration Memory Device option so the design boots automatically from flash on every power-up.

11.3 Run Instructions

1. Power the FPGA board; the design auto-loads from the onboard SPI flash.
2. Press the reset switch to initialize the system to the IDLE state.
3. Enter the 4-digit password (1234) using the hexadecimal keypad to gain access.
4. Observe the seven-segment display and status LEDs for the authentication result ("PASS" or "FAIL").
5. Entering any value other than 1234 displays "FAIL" and increments the failed-attempt counter.
6. After 3 consecutive incorrect attempts, the display shows "LOCK" and the system stops accepting further password entries.
7. Press key "A" on the keypad to unlock the system and return it to IDLE.

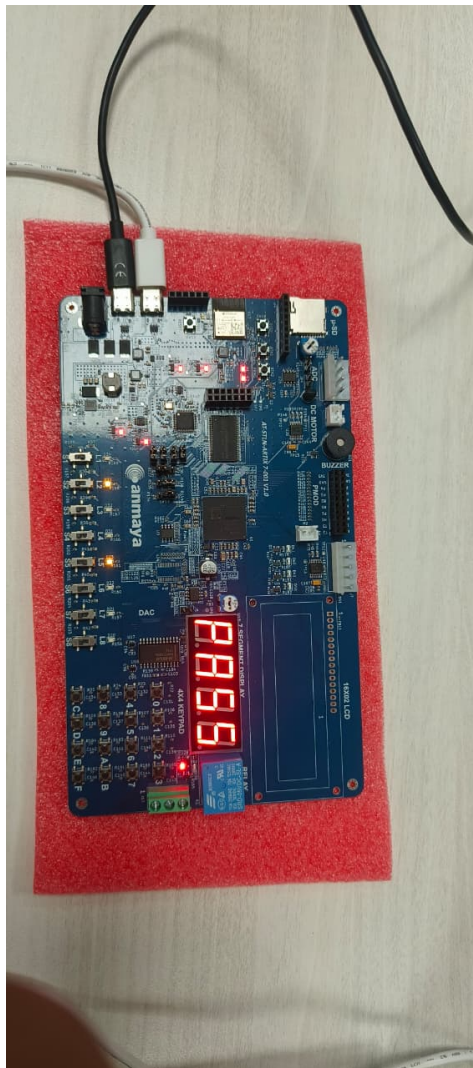
11.4 Testing Procedure

Test Case	Description	Expected Result	Actual Result	Status
TC-01	Correct Password Authentication (1234)	Seven-segment display shows "PASS", status LEDs indicate grant/match, and failed-attempt counter resets to zero.	System operated as expected.	Pass
TC-02	Incorrect Password Authentication	Seven-segment display shows "FAIL", status LEDs indicate deny, and failed-attempt counter increments by one.	System operated as expected.	Pass
TC-03	Security Lockout after Three Failed Attempts	After the third consecutive incorrect password, the FSM transitions to the LOCK state, the seven-segment display shows "LOCK", and further password entries are ignored.	System operated as expected.	Pass
TC-04	Unlock from LOCK State	Pressing the designated unlock key while in the LOCK state resets the failed-attempt counter and returns the FSM to the IDLE state.	System operated as expected.	Pass
TC-05	XOR Password Encryption Check	The stored password is correctly encrypted and decrypted using the XOR key, and only the correct 4-digit password results in a match.	Encryption and comparison verified successfully.	Pass

12. Test Results

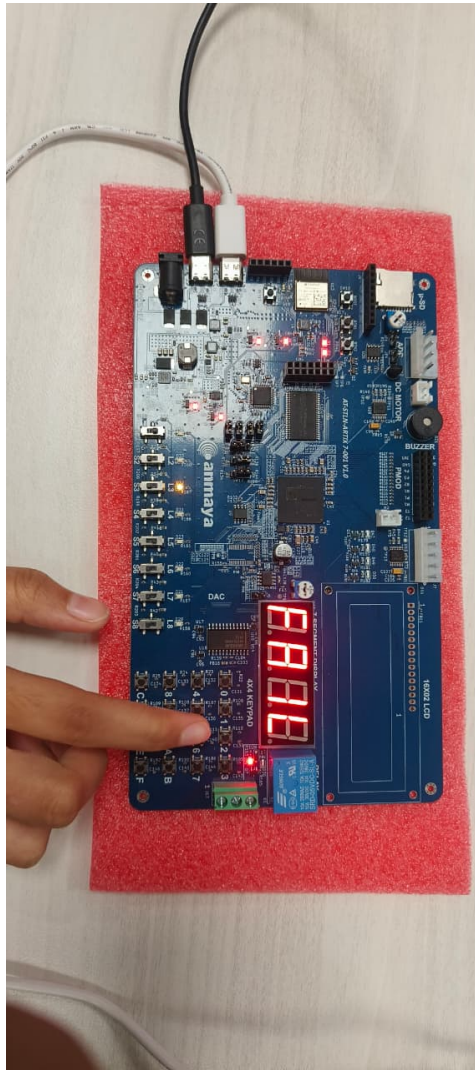
The FPGA-based system successfully authenticated users using keypad input. Entering the correct password (1234) resulted in a PASS indication on the seven-segment display. Entering any incorrect password resulted in a FAIL indication on the seven-segment display, and three consecutive incorrect attempts correctly triggered the LOCK state, with the display showing "LOCK" and all further entries blocked until key "A" was pressed to unlock the system. The system successfully demonstrated FPGA-based keypad scanning, XOR password encryption, finite state machine control, and seven-segment display driving, with the design itself running from the onboard SPI configuration flash.

Figure 12.1: Correct Password — "PASS"



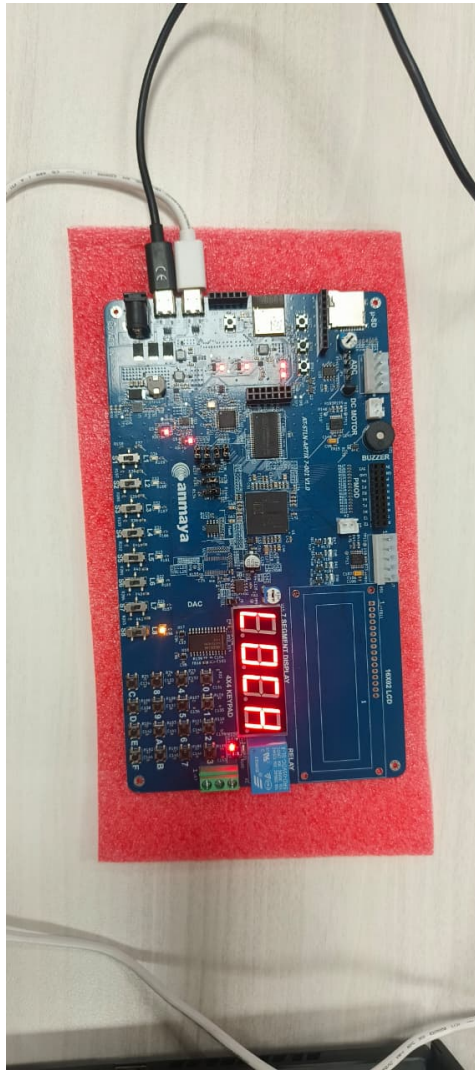
Entering the correct password 1234 on the hexadecimal keypad causes the security FSM to grant access; the MAX7219 seven-segment display reads "PASS".

Figure 12.2: Incorrect Password — "FAIL"



Entering any password other than 1234 causes the security FSM to deny access; the seven-segment display reads "FAIL" and the failed-attempt counter is incremented.

Figure 12.3: Security Lockout — "LOCK"



After three consecutive incorrect password attempts, the security FSM transitions to the LOCK state; the seven-segment display reads "LOCK" and further keypad entries are ignored until key "A" is pressed to unlock the system.

Functional simulation in Vivado confirmed correct behaviour of the keypad decoder, password entry register, password comparator, security FSM and display driver across the PASS, FAIL and LOCK conditions, matching the results observed on hardware.

13. Conclusion

The FPGA-Based Digital Password Authentication System with Security Lockout Mechanism successfully combines keypad-based password entry, XOR-based password encryption, finite state machine controlled PASS/FAIL/LOCK logic, and seven-segment display feedback into a reliable digital access-control solution. The project demonstrates effective FPGA hardware design using Verilog HDL and highlights how a security lockout mechanism can significantly improve resistance against repeated unauthorized access attempts compared to conventional password-based systems.

14. References

- Xilinx Vivado Design Suite Documentation
- Artix-7 FPGA User Manual
- AT-STLN-ARTIX 7-001 Hardware Reference Manual, Anmaya Technologies (ANM-PRD-2025-005)
- MAX7219 Seven-Segment Display Driver Datasheet
- Verilog HDL Reference Manual